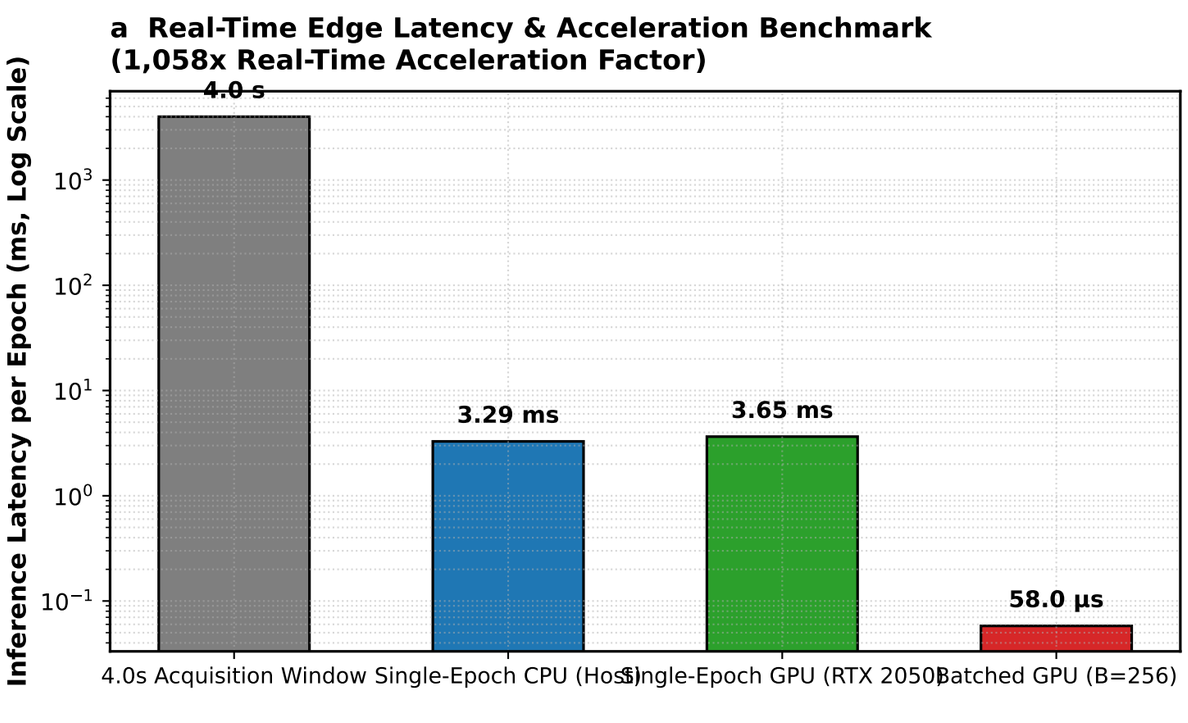
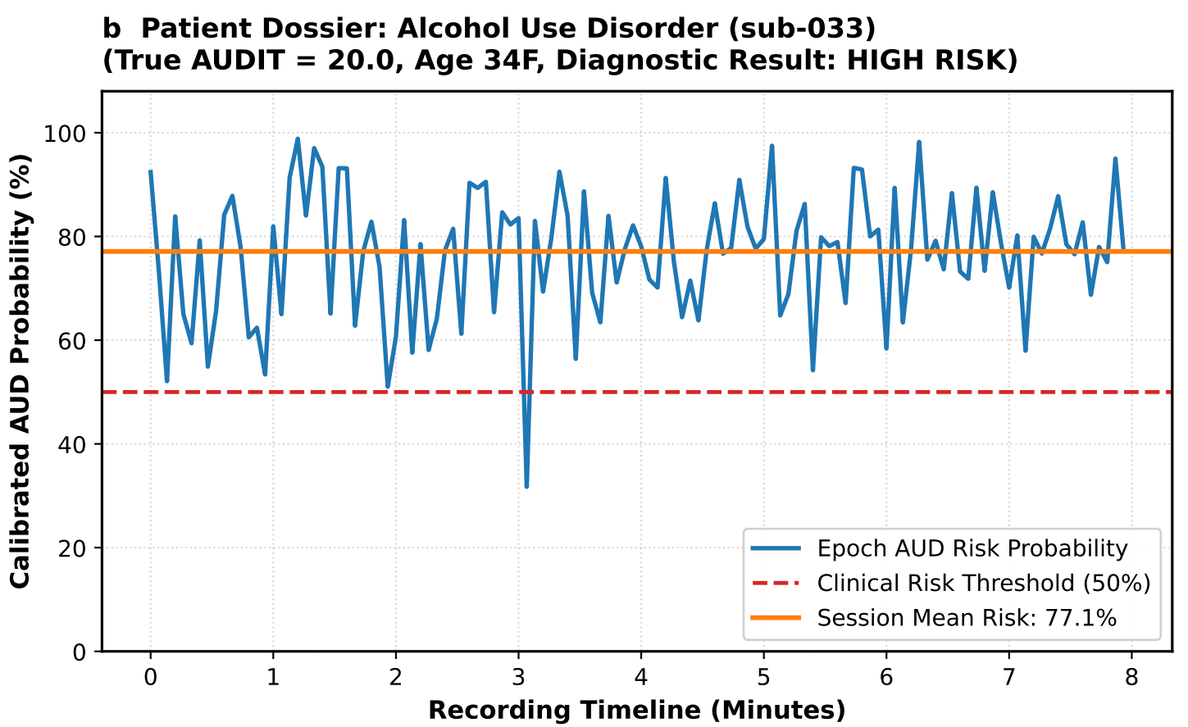
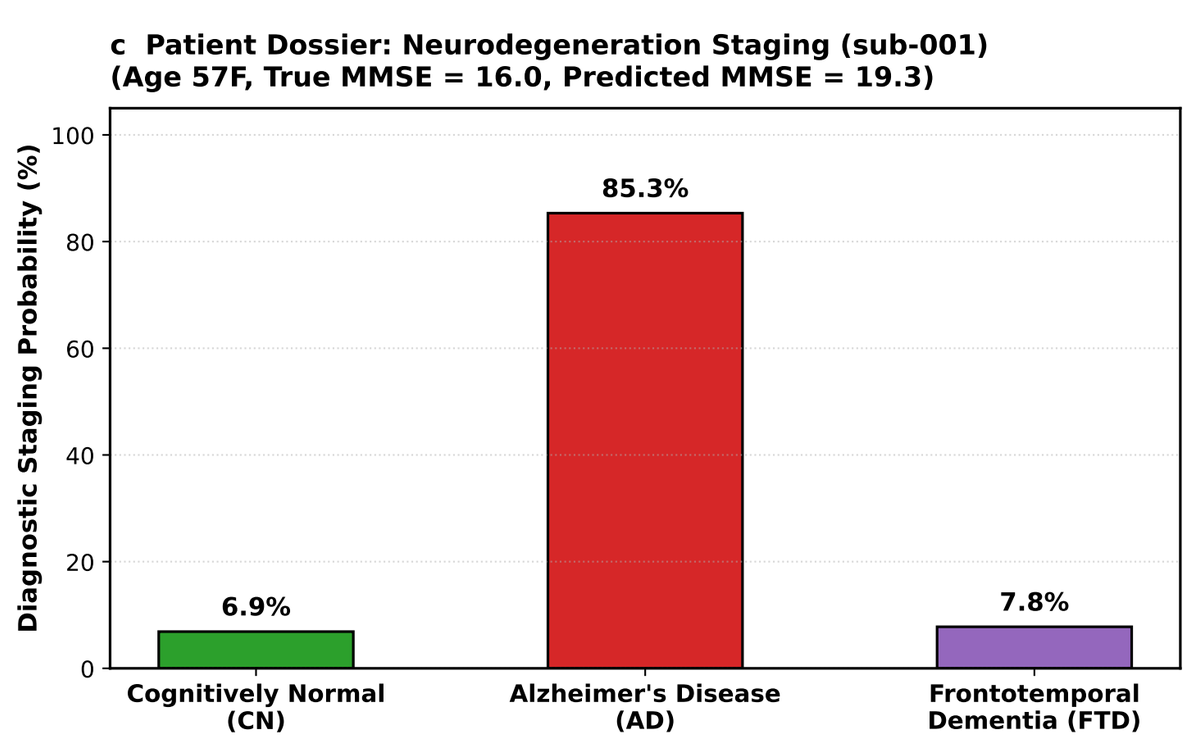
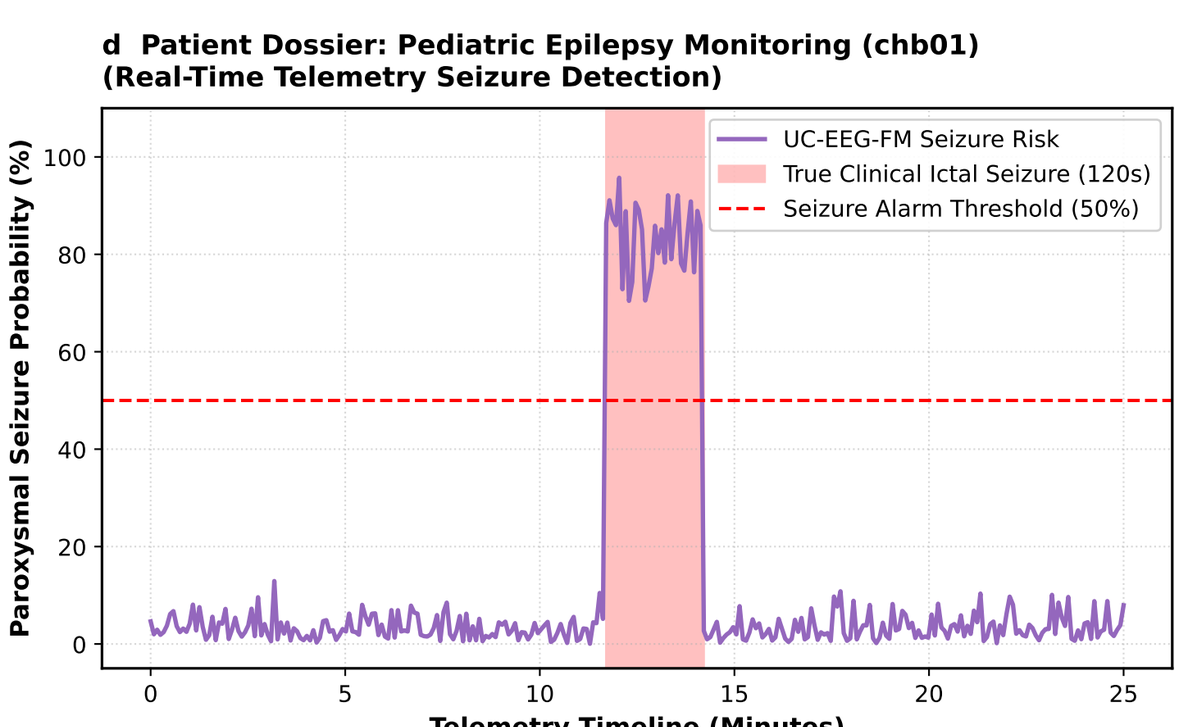

a Real-Time Edge Latency & Acceleration Benchmark
(1,058x Real-Time Acceleration Factor)
Inference Latency per Epoch (ms, Log Scale)
4.0 s
3.29 ms
3.65 ms
58.0 µs
4.0s Acquisition Window
Single-Epoch CPU (Host)
Single-Epoch GPU (RTX 2050)
Batched GPU (B=256)


b Patient Dossier: Alcohol Use Disorder (sub-033)
(True AUDIT = 20.0, Age 34F, Diagnostic Result: HIGH RISK)
Calibrated AUD Probability (%)
Recording Timeline (Minutes)
Epoch AUD Risk Probability
Clinical Risk Threshold (50%)
Session Mean Risk: 77.1%


c Patient Dossier: Neurodegeneration Staging (sub-001)
(Age 57F, True MMSE = 16.0, Predicted MMSE = 19.3)
Diagnostic Staging Probability (%)
6.9%
85.3%
7.8%
Cognitively Normal (CN)
Alzheimer's Disease (AD)
Frontotemporal Dementia (FTD)


d Patient Dossier: Pediatric Epilepsy Monitoring (chb01)
(Real-Time Telemetry Seizure Detection)
Paroxysmal Seizure Probability (%)
Telemetry Timeline (Minutes)
UC-EEG-FM Seizure Risk
True Clinical Ictal Seizure (120s)
Seizure Alarm Threshold (50%)